Appendix for "Blending Proxy Metrics with a North Star"

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A $\underline{n}(\tilde{A})$ is a Conservative Bound

Let $\underline{n}(A) := (z^2 - 1)\frac{2A^\top \Omega A}{A^\top \Sigma_X A}$. The goal is to show that $\underline{n}(\tilde{A}) \geq \underline{n}(A^*)$, and so checking that $n \geq \underline{n}(\tilde{A})$ is sufficient to establish that $n \geq \underline{n}(A^*)$, which is a prerequisite to using the closed-form approximation to the optimal weights.

First, under joint normality of $X = (P, Y)$, the expected returns to a single experiment for any choice of A are given by

$$R(A) = \frac{\Gamma^\top A}{\sqrt{A^\top \Sigma_{\widehat{X}} A}} \phi\left(z\sqrt{q_n(A)}\right),$$

where $q_n(A) = \frac{A^\top (2\Omega/n) A}{A^\top \Sigma_{\widehat{X}} A}$.

Note that the returns are invariant to the scale of A and depend only on its direction. Therefore, without loss of generality, I can fix the scale of A by requiring $\Gamma^\top A = 1$. Under this normalization, maximizing $R(A)$ is equivalent to minimizing the following objective function:

$$F(A) := \log A^\top \Sigma_{\widehat{X}} A + z^2 q_n(A).$$

Abusing notation, I will redefine A^* as the solution to the following constrained minimization problem:

$$A^* = \arg \min_{\Gamma^\top A = 1} F(A). \tag{1}$$

Also under this normalization, $\tilde{A}$ can be written as the solution to the following constrained minimization problem:

$$\tilde{A} = \arg \min_{\Gamma^\top A = 1} \underbrace{A^\top (\Sigma_X + (1 + z^2) 2\Omega/n) A}_{W(A)}. \tag{2}$$

The following assumes that both A^* and $\tilde{A}$ are interior solutions to their respective minimization problems, i.e., $A_i^* > 0$ and $\tilde{A}_i > 0$ for all i.

$W(A)$ can be rewritten as

$$\begin{aligned} W(A) &= A^\top(\Sigma_X + (1+z^2)2\Omega/n)A & (3)\\ &= A^\top \Sigma_{\widehat{X}} A\left(1 + z^2 \frac{A^\top(2\Omega/n)A}{A^\top \Sigma_{\widehat{X}} A}\right) & (4)\\ &= v_A(1 + z^2 q_n(A)), & (5)\end{aligned}$$

where $v_A := A^\top \Sigma_{\widehat{X}} A$.

Since $\tilde{A}$ minimizes $W(A)$, we have that:

$$\begin{aligned} v_{\tilde{A}}(1 + z^2 q_n(\tilde{A})) &\leq v_{A^*}(1 + z^2 q_n(A^*)) & (6)\\ \frac{v_{\tilde{A}}}{v_{A^*}} &\leq \frac{1 + z^2 q_n(A^*)}{1 + z^2 q_n(\tilde{A})}. & (7)\end{aligned}$$

Now consider $F(\tilde{A}) - F(A^*)$:

$$F(\tilde{A}) - F(A^*) = \log \frac{v_{\tilde{A}}}{v_{A^*}} + z^2(q_n(\tilde{A}) - q_n(A^*)). \quad (8)$$

(7) implies that $F(\tilde{A}) - F(A^*)$ is bounded above by

$$F(\tilde{A}) - F(A^*) \leq \log \frac{1 + z^2 q_n(A^*)}{1 + z^2 q_n(\tilde{A})} + z^2(q_n(\tilde{A}) - q_n(A^*)). \quad (9)$$

Grouping terms, we have that

$$F(\tilde{A}) - F(A^*) \leq [\log(1 + z^2 q_n(A^*)) - z^2 q_n(A^*)] - [\log(1 + z^2 q_n(\tilde{A})) - z^2 q_n(\tilde{A})]. \quad (10)$$

Suppose, by way of contradiction, that $q_n(A^*) > q_n(\tilde{A})$. Because $\log(1 + x) - x$ is strictly decreasing on the domain of $q_n(A)$ (i.e., for $x > 0$), $q_n(A^*) > q_n(\tilde{A})$ implies that the term on the right-hand side of the inequality is negative. However, $F(\tilde{A}) - F(A^*)$ cannot be negative, since A^* minimizes $F(A)$. Therefore, it must be the case that $q_n(A^*) \leq q_n(\tilde{A})$. $q_n(A)$ is an increasing function of $\frac{A^\top \Omega A}{A^\top \Sigma_X A}$, so this implies that

$$\frac{A^{*\top} \Omega A^*}{A^{*\top} \Sigma_X A^*} \leq \frac{\tilde{A}^\top \Omega \tilde{A}}{\tilde{A}^\top \Sigma_X \tilde{A}}. \quad (11)$$

Multiplying both sides by $2(z^2 - 1)$ yields

$$\underline{n}(A^*) \leq \underline{n}(\tilde{A}), \quad (12)$$

as desired.

B Average Returns are Decreasing in n for $z \leq 2.414$

The expected returns of a single experiment for any choice of A are given by:

$$R(A,n) = \frac{\Gamma^\top A}{\sqrt{A^\top \Sigma_{\widehat{X}} A}} \phi\left(z\sqrt{q_n(A)}\right), \tag{13}$$

where $q_n(A) := \frac{A^\top (2\Omega/n) A}{A^\top \Sigma_{\widehat{X}} A}$ is the ratio of the within-experiment sampling variance of $\widehat{Z}(A)$ to its total variance across experiments, $\Sigma_{\widehat{X}} = \Sigma_X + 2\Omega/n$.

Note that

$$\begin{aligned} q_n(A) = q_A(n) &= \frac{A^\top (2\Omega/n) A}{A^\top \Sigma_{\widehat{X}} A} &(14)\\ &= \frac{A^\top (2\Omega) A}{n A^\top \Sigma_X A + A^\top (2\Omega) A}, &(15) \end{aligned}$$

so $q_A(n)$ is strictly decreasing in n.

Isolating the terms that depend on n, we have that:

$$R(A,n)/n \propto \underbrace{\frac{q_A(n)}{\sqrt{1-q_A(n)}} \phi\left(z\sqrt{q_A(n)}\right)}_{F_z(q)}, \tag{16}$$

for $q_A(n) \in (0,1)$.

Taking logs and differentiating with respect to q, we have:

$$\frac{d}{dq} \log F_z(q) = \frac{1}{q} + \frac{1}{2(1-q)} - \frac{z^2}{2}. \tag{17}$$

The first two terms are minimized at $q^* = 2 - \sqrt{2} \in (0,1)$ and the minimum value is $1 + \frac{\sqrt{2}}{2} + \frac{1+\sqrt{2}}{2} = \frac{3}{2} + \sqrt{2} > 0$. Therefore, the derivative remains positive so long as $z^2 \leq 3 + 2\sqrt{2}$. Set $\sqrt{3+2\sqrt{2}} = \sqrt{a} + \sqrt{b}$.

Squaring both sides of the inequality, we have $3 + 2\sqrt{2} = a + b + 2\sqrt{ab}$. If we set $a + b = 3$ and $2\sqrt{ab} = 2\sqrt{2}$, we have $a = 2$ and $b = 1$. Therefore, R/n is increasing in q as long as $1 \leq z \leq 1 + \sqrt{2} \approx 2.414$. Since q is decreasing in n, this implies that R/n is decreasing in n for the same range of z values.

In other words, the average return per unit of allocation is decreasing in n as long as the launch threshold is not too strict. If it is very strict, then almost no treatment will be launched unless the measurement is sufficiently precise.